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United States Patent	12387949
Kind Code	B2
Date of Patent	August 12, 2025
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Substrate processing apparatus

Abstract

Described herein is a technique capable of reducing an amount of moisture in a low temperature region in a substrate processing apparatus provided with a transfer chamber. According to one aspect of the technique, there is provided a substrate processing apparatus including: a process chamber provided with a heater; a load lock chamber; a transfer chamber provided between the process chamber and the load lock chamber and including a first region provided adjacent to the process chamber and a second region provided more adjacent to the load lock chamber than the first region and whose temperature is lower than a temperature of the first region; a detector capable of detecting an amount of moisture in the transfer chamber; and an inert gas supplier capable of supplying an inert gas toward the second region in the transfer chamber.

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Appl. No.: 18/612702

Filed: March 21, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240234183 A1	Jul. 11, 2024

Foreign Application Priority Data

JP	2021-006865	Jan. 20, 2021
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Related U.S. Application Data

continuation parent-doc US 17205580 20210318 US 11967513 child-doc US 18612702

Publication Classification

Int. Cl.: **H01L21/67** (20060101); **F27B17/00** (20060101); **F27D3/00** (20060101); F27D9/00 (20060101)

U.S. Cl.:

CPC **H01L21/67196** (20130101); **F27B17/0025** (20130101); **F27D3/0084** (20130101); **H01L21/67103** (20130101); **H01L21/67167** (20130101); **H01L21/67201** (20130101); **H01L21/67242** (20130101); F27D2009/0013 (20130101)

Field of Classification Search

CPC: H01L (21/67196); H01L (21/67103); H01L (21/67167); H01L (21/67201); H01L (21/67242); F27B (17/0025); F27D (3/0084); F27D (2009/0013)

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This non-provisional U.S. patent application is a continuation of and claims priority to U.S. patent application Ser. No. 17/205,580, filed Mar. 18, 2021, which claims priority under 35 U.S.C. § 119 of Japanese Patent Application No. 2021-006865, filed on Jan. 20, 2021, in the Japanese Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Field

(1) The present disclosure relates to a substrate processing apparatus.

2. Related Art

(2) As an apparatus of manufacturing a semiconductor device, an apparatus including a process chamber in which a substrate is processed and a transfer chamber in which a robot capable of transferring the substrate is provided may be used. When a substance unrelated to a substrate processing is present in the transfer chamber (for example, moisture is present in the transfer chamber), a yield may be reduced. Therefore, an amount of a foreign matter such as the moisture in the transfer chamber should be reduced. According to some related arts, the entirety of the transfer chamber is heated to remove the moisture.

(3) The moisture may be abundant in a low temperature region of the transfer chamber. Then, when trying to process the transfer chamber as described above (for example, by heating the entirety of transfer chamber), the moisture may not be completely removed.

SUMMARY

(4) Described herein is a technique capable of reducing an amount of moisture in a low temperature region in a substrate processing apparatus provided with a transfer chamber.

(5) According to one aspect of the technique of the present disclosure, there is provided a substrate processing apparatus including: a process chamber provided with a heater; a load lock chamber; a transfer chamber provided between the process chamber and the load lock chamber and including a first region provided adjacent to the process chamber and a second region provided more adjacent to the load lock chamber than the first region and whose temperature is lower than a temperature of the first region; and an inert gas supplier capable of supplying an inert gas toward the second region in the transfer chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically illustrates a horizontal cross-section of a substrate processing apparatus according to a first embodiment described herein.

(2) FIG. 2 schematically illustrates a vertical cross-section of the substrate processing apparatus according to the first embodiment described herein.

(3) FIGS. 3A and 3B schematically illustrate a distributor of the substrate processing apparatus according to the first embodiment described herein.

(4) FIG. 4 schematically illustrates a vertical cross-section of a reactor (RC) of the substrate processing apparatus according to the first embodiment described herein.

(5) FIG. 5 schematically illustrates a gas supplier of the substrate processing apparatus according to the first embodiment described herein.

(6) FIG. 6 schematically illustrates a configuration of a controller of the substrate processing apparatus and related components of the substrate processing apparatus according to the first embodiment described herein.

(7) FIG. 7 schematically illustrates a horizontal cross-section of a substrate processing apparatus according to a second embodiment described herein.

(8) FIG. 8 schematically illustrates a vertical cross-section of the substrate processing apparatus according to the second embodiment described herein.

(9) FIG. 9 schematically illustrates a horizontal cross-section of a substrate processing apparatus according to a third embodiment described herein.

(10) FIG. 10 schematically illustrates a horizontal cross-section of a substrate processing apparatus according to a fourth embodiment described herein.

(11) FIG. 11 schematically illustrates a vertical cross-section of the substrate processing apparatus according to the fourth embodiment described herein.

DETAILED DESCRIPTION

Embodiments

(12) Hereinafter, one or more embodiments (also simply referred to as “embodiments”) according to the technique of the present disclosure will be described with reference to the drawings.

First Embodiment

(13) Hereinafter, a first embodiment according to the technique of the present disclosure will be described.

(14) (1) Configuration of Substrate Processing Apparatus

(15) A substrate processing apparatus according to the first embodiment will be described with reference to FIGS. 1 through 6. The drawings used in the following descriptions are all schematic. For example, a relationship between dimensions of each component and a ratio of each component shown in the drawings may not always match the actual ones. Further, even between the drawings, the relationship between the dimensions of each component and the ratio of each component may not always match.

(16) FIGS. 1 and 2 schematically illustrate the substrate processing apparatus according to the first embodiment, and FIGS. 3A and 3B schematically illustrate a distributor of an inert gas supplier provided in a transfer chamber of the substrate processing apparatus according to the first embodiment. FIGS. 4 and 5 schematically illustrate a reactor (RC) of the substrate processing apparatus according to the first embodiment. FIG. 6 schematically illustrates a configuration of a controller of the substrate processing apparatus and related components of the substrate processing apparatus according to the first embodiment. Each configuration will be described below in detail.

(17) The configuration of the substrate processing apparatus will be described with reference to FIGS. 1 and 2. FIG. 1 schematically illustrates a horizontal cross-section of an exemplary configuration of the substrate processing apparatus. FIG. 2 schematically illustrates a vertical cross-section of the exemplary configuration of the substrate processing apparatus taken along the line α - α' in FIG. 1.

(18) A substrate processing apparatus **200** according to the first embodiment is configured to process a plurality of substrates including a substrate **100**. Hereinafter, the plurality of the substrates including the substrate **100** may also be simply referred to as substrates **100**. The substrate processing apparatus **200** includes an I/O stage (input/output stage) **110**, an atmospheric transfer chamber **120**, a load lock chamber **130**, a vacuum transfer chamber **140** and a plurality of process modules (hereinafter, also simply referred to as “PMs”) such as process modules PM1, PM2, PM3 and PM4. Hereinafter, one of the process modules PM1, PM2, PM3 and PM4 may be referred to as a “process module PM”, and the process modules PM1, PM2, PM3 and PM4 may be collectively referred to as “process modules PMs”.

(19) <Atmospheric Transfer Chamber and I/O Stage>

(20) The I/O stage (also referred to as a “loading port shelf”) **110** is provided in front of the

substrate processing apparatus **200**. The I/O stage **110** is configured such that a plurality of pods including a pod **111** can be placed on the I/O stage **110**. Hereinafter, the plurality of the pods including the pod **111** may also be simply referred to as pods **111**. The pod **111** is used as a carrier for transferring the substrate (wafer) **100** such as a silicon (Si) substrate.

(21) The substrates **100** managed on a lot basis can be stored in the pod **111**. For example, n substrates (n is a natural number) are stored in the pod **111** as the substrates **100**.

(22) A cap **112** is installed at the pod **111**. The cap **112** is opened or closed by a pod opener **121**. The pod opener **121** is configured to open and close the cap **112** of the pod **111** placed on the I/O stage **110**. When the pod opener **121** opens a substrate entrance (not shown) of the pod **111**, the substrates **100** may be loaded into or unloaded out of the pod **111**. The pod **111** is provided to or discharged from the I/O stage **110** by an automated material handling systems (AMHS) (not shown).

(23) The I/O stage **110** is provided adjacent to the atmospheric transfer chamber **120**. The load lock chamber **130**, which will be described later, is connected to a side of the atmospheric transfer chamber **120** other than a side at which the I/O stage **110** is provided. An atmospheric transfer robot **122** capable of transferring the substrates **100** is provided in the atmospheric transfer chamber **120**.

(24) A communication hole **128** configured to transfer the substrates **100** into or out of the atmospheric transfer chamber **120** and the pod opener **121** are provided at a front side of a housing **127** of the atmospheric transfer chamber **120**. A communication hole **129** configured to transfer the substrates **100** into or out of the load lock chamber **130** is provided at a rear side of the housing **127** of the atmospheric transfer chamber **120**. When the communication hole **129** is opened by a gate valve **133**, the substrates **100** may be loaded into the load lock chamber **130** or unloaded out of the load lock chamber **130**.

(25) <Load Lock Chamber>

(26) The load lock chamber **130** is provided adjacent to the atmospheric transfer chamber **120**. The vacuum transfer chamber **140**, which will be described later, is provided at a side of a housing **131** constituting the load lock chamber **130** other than a side of the housing **131** that is adjacent to the atmospheric transfer chamber **120**.

(27) A substrate mounting table **136** provided with at least two placing surfaces **135** is provided in the load lock chamber **130**. A distance between the two placing surfaces **135** may be set based on a distance between end effectors of an arm of a robot **170** which will be described later.

(28) <Vacuum Transfer Chamber>

(29) The substrate processing apparatus **200** includes the vacuum transfer chamber (also referred to as a “transfer module”) **140**, that is, a transfer space in which the substrates **100** are transferred under a negative pressure. The vacuum transfer chamber **140** may also be simply referred to as a “transfer chamber **140**”. For example, a housing **141** constituting the vacuum transfer chamber **140** is pentagonal when viewed from above. The load lock chamber **130** and the process modules PM1, PM2, PM3 and PM4 where the substrates **100** are processed are connected to respective sides of the housing **141** of a pentagonal shape. The robot **170** capable of transferring the substrates **100** under the negative pressure is provided at approximately at a center of the vacuum transfer chamber **140** with a flange **144** as a base. The robot **170** serves as a transfer device.

(30) The load lock chamber **130** and the vacuum transfer chamber **140** are communicated with each other through a communication hole **142**. The communication hole **142** is opened or closed by a gate valve **134**.

(31) The robot **170** provided in the vacuum transfer chamber **140** may be elevated and lowered by an elevator **145** while maintaining the vacuum transfer chamber **140** airtight by the flange **144**. The elevator **145** is configured to elevate and lower two arms including an arm **172** of the robot **170**. In FIG. 2, for convenience of explanation, the end effectors of the arm **172** are illustrated, and components such as a link structure between the end effectors and the flange **144** are omitted.

(32) The reactor (hereinafter, also referred to as an “RC”) are provided in each of the process modules PM1, PM2, PM3 and PM4 adjacent to the vacuum transfer chamber 140. Specifically, reactors RC1 and RC2 are provided in the process module PM1. Reactors RC3 and RC4 are provided in the process module PM2. Reactors RC5 and RC6 are provided in the process module PM3. Reactors RC7 and RC8 are provided in the process module PM4. Hereinafter, one of the reactors RC1 through RC8 may be referred to as a “reactor RC”, and the reactors RC1 through RC8 may be collectively referred to as “reactors RCs”.

(33) A communication hole such as a communication hole 148 shown in FIG. 4 is provided in each of sidewalls of the housing 141 facing the reactors (RCs) RC1 through RC8, respectively. For example, a communication hole 148-5 is provided in the sidewall of the housing 141 facing the reactor RC5 as shown in FIG. 2. A gate valve such as a gate valve 149 shown in FIG. 4 is provided in each of the reactors (RCs) RC1 through RC8. For example, a gate valve 149-5 is provided in the sidewall of the housing 141 facing the reactor RC5 as shown in FIG. 2. Hereinafter, the communication hole including the communication hole 148 may also be collectively or individually referred to as the communication hole 148, and the gate valve including the gate valve 149 may also be collectively or individually referred to as the gate valve 149. Since the configurations of the reactors RC1 through RC4 and RC6 through RC8 are the same as that of the reactor RC5, the detailed description thereof will be omitted.

(34) An arm controller 171 capable of controlling an elevating operation and a rotating operation of the arm 172 is embedded in the elevator 145. The arm controller 171 mainly includes a support shaft 171a configured to support a shaft of the arm 172 and an actuator 171b configured to elevate or rotate the support shaft 171a.

(35) The actuator 171b may include an elevator 171c such as a motor configured to elevate and lower the support shaft 171a and a rotator 171d such as a gear configured to rotate the support shaft 171a. The elevator 145 may further include an instruction controller 171e which is a part of the arm controller 171 and configured to control the actuator 171b to move the support shaft 171a up and down or to rotate the support shaft 171a. The instruction controller 171e may be electrically connected to a controller 400 described later. The actuator 171b may be controlled by the instruction controller 171e based on an instruction from the controller 400.

(36) The arm 172 can be rotated and stretched about the shaft. As described above, a shaft of the robot 170 is arranged approximately at a center of the housing 141. However, a distance from a center of the shaft to a substrate mounting table 212 (described later) of each of the reactors (RCs) may differ due to a structural restriction. For example, in FIG. 1, a distance L1 from the center of the shaft of the robot 170 to the substrate mounting table 212 of the reactor RC8 (or RC7) is shorter than a distance L2 from the center of the shaft of the robot 170 to the substrate mounting table 212 of the reactor RC4 (or RC3).

(37) By rotating and stretching the robot 170, the substrate 100 can be transferred into or out of the reactors RCs. As described above, distances between each of the reactors RCs and the shaft of the robot 170 are different for each of the reactors RCs. The robot 170 can transfer the substrate 100 to the reactor RC, for example, in accordance with an instruction from the controller 400.

(38) Subsequently, an exhaustor (which is an exhaust system) 160 will be described. The exhaustor 160 is provided below the housing 141. Specifically, for example, an exhaust pipe 161 is connected to a bottom wall of the housing 141. An APC (Automatic Pressure Controller) 162 is provided at the exhaust pipe 161. The APC 162 serves as a pressure controller (pressure regulator) capable of controlling an inner atmosphere of the housing 141 to a predetermined pressure. The APC 162 includes a valve body (not shown) whose opening degree can be adjusted. The APC 162 can adjust the conductance of the exhaust pipe 161 in accordance with an instruction from the controller 400. Further, a valve 163 is provided at the exhaust pipe 161. The exhaust pipe 161, the APC 162 and the valve 163 may be collectively referred to as a “transfer chamber exhaustor”.

(39) Further, a dry pump DP (not shown) is provided at a downstream side of the exhaust pipe 161.

The dry pump is capable of exhausting the inner atmosphere of the housing **141** through the exhaust pipe **161**.

(40) A moisture detector **146** is provided in the housing **141** constituting the vacuum transfer chamber **140**. The moisture detector **146** is electrically connected to the controller **400**. The moisture detector **146** is capable of detecting an amount of moisture (water vapor) in the vacuum transfer chamber **140** and transmitting the detected amount of the moisture to the controller **400**. The moisture detector **146** may also be simply referred to as a “detector”.

(41) For the reason described later, the moisture detector **146** is provided at a location at which the amount of the moisture in the vacuum transfer chamber **140** can be detected. In the present specification, the “location at which the amount of the moisture in the vacuum transfer chamber **140** can be detected” may refer to a low temperature location, for example, the vicinity of a ceiling **147** of the housing **141** and the vicinity of a side wall **141a** (described later) provided adjacent to the load lock chamber **130**.

(42) A window **151** is provided on the ceiling **147**. The window **151** is used to confirm whether or not an operation of the robot **170** is normal. An O-ring **152** serving as a seal is arranged between the window **151** and a wall **147a** constituting the ceiling **147**. For example The O-ring **152** is made of a material such as rubber. As a result, the inner atmosphere of the vacuum transfer chamber **140** is sealed. A lid **153** is provided on the window **151**.

(43) A flow path **154** configured to flow cooling water or a chiller (medium) capable of adjusting a temperature of the housing **141** may be provided at the housing **141**. With such a structure, even when the housing **141** is affected by a heater **213** (see FIG. 4) in the reactor RC, it is possible to suppress an excessive temperature elevation.

(44) An inert gas supplier (which is an inert gas supply system) **180** capable of supplying an inert gas to the low temperature location described later is provided at the housing **141**. For example, as shown in FIG. 2, the inert gas supplier **180** may be provided on the ceiling **147**. The inert gas supplier **180** includes an inert gas supply pipe **181**. An inert gas supply source **182**, a mass flow controller (MFC) **183** serving as a flow rate controller (flow rate control structure) and a valve **184** serving as an opening/closing valve are sequentially provided in order at the inert gas supply pipe **181** from an upstream side toward a downstream side of the inert gas supply pipe **181**. A heater **185** capable of heating the inert gas supplied in the inert gas supply pipe **181** may be further provided.

(45) A distributor **186** is provided at a tip (front end) of the inert gas supply pipe **181**. The distributor **186** is configured to disperse and supply the inert gas into the housing **141**.

(46) The inert gas supplier **180** is constituted mainly by the inert gas supply pipe **181**, the MFC **183**, the valve **184** and the distributor **186**. The inert gas supplier **180** may further include the heater **185**. Since the inert gas supplier **180** is configured to supply the inert gas to the transfer chamber **140**, the inert gas supplier **180** may also be referred to as a “transfer chamber inert gas supplier”.

(47) Subsequently, the low temperature location and a high temperature location will be described. For example, the high temperature location may refer to walls such as a wall **141b** provided adjacent to the reactor RC5 shown in FIG. 2. When processing the substrates **100**, the substrates **100** are heated by heaters such as the heater **213** in the reactor RC shown in FIG. 4. Therefore, the walls adjacent to the reactors RC such as the wall **141b** are affected by the heaters such as the heater **213**, and temperatures of the walls are higher than that of the side wall (also simply referred to as a wall) **141a** provided adjacent to the load lock chamber **130**. In the present specification, the high temperature location may refer to a location that becomes hot due to an influence of the heaters of the reactors RCs such as the heater **213**. A region including the high temperature location may also be referred to as a “high temperature region” or a “first region”.

(48) In the present specification, the low temperature location may refer to a location whose temperature is lower than that of the high temperature location. A region including the low temperature location may also be referred to as a “low temperature region” or a “second region”.

For example, the low temperature location may refer to the ceiling **147** or the wall **141a** of the transfer chamber **140** provided with the communication hole **142**, and the low temperature region may refer to the region in which the ceiling **147** or the wall **141a** of the transfer chamber **140** is provided. A region in which the O-ring **152** is arranged may also be referred to as the low temperature region. Since the components described above such as the ceiling **147**, the wall **141a** and the O-ring **152** are located far from the reactors RCs, they are not easily affected by the heaters provided in the reactors RCs such as the heater **213**. Therefore, temperatures of the components described above are lower than that of the wall **141b**. Further, since an outer shell of the low temperature location such as the ceiling **147** is exposed to an outer atmosphere, a temperature of the outer shell of the low temperature location may become close to the room temperature at which the moisture easily adheres. That is, the moisture may easily adhere to the outer shell of the low temperature location.

(49) It can be said that the high temperature location is provided between the heaters in the reactors RCs (such as the heater **213**) and the low temperature location. It can be said that the low temperature location is provided between the high temperature location and the load lock chamber **130**. According to the present embodiment, the “low temperature” may refer to a temperature (for example, less than 100° C.) low enough to allow the moisture to adhere in the transfer chamber **140**.

(50) For example, a central portion of the ceiling **147** in the horizontal direction is separated from each of the reactors RCs. Therefore, the central portion of the ceiling **147** is not easily affected by the heat of the heaters such as the heater **213**. Similarly, the vicinity of the communication hole **142** is not easily affected by the heat of the heaters such as the heater **213**. Therefore, temperatures of the central portion of the ceiling **147** and the vicinity of the communication hole **142** are low.

(51) Further, when the chiller is supplied, the housing **141** is maintained at a temperature (for example, the room temperature) at which the housing **141** can be operated by, for example, a maintenance personnel. Therefore, since the temperature of the housing **141** is stable and low, the moisture is more likely to adhere to the low temperature location.

(52) There is a problem that an amount of the moisture increases in such a low temperature location. The moisture adhering to the low temperature location may adhere to the substrate **100**, particularly the processed substrate **100** heated by a substrate processing. Thereby, a natural oxide film may be formed on the substrate **100**, and the substrate **100** may be unintentionally modified by components (such as hydrogen (H) and oxygen (O)) of the moisture.

(53) As described above, the distance between each of the reactors RCs and the shaft of the robot **170** is different for each of the reactors RCs. Then, transfer distances of the substrates **100** may also be different for each of the substrates **100**. Therefore, a state of each of the substrates **100** (for example, the formation of the natural oxide film and the unintended modification of a film formed on each of the substrates **100**) may be different for each of the substrates **100** depending on the reactor RC in which the substrate **100** is processed. Therefore, a yield may be reduced.

(54) In order to address the problem described above, the housing **141** may be heated to remove the moisture. However, when the housing **141** is heated, for example, the O-ring **152** in the ceiling **147** and components constituting the robot **170** may be deteriorated by heating the housing **141**. Therefore, it is not preferable to heat the housing **141** (that is, the transfer chamber **140**).

(55) Therefore, according to the present embodiment, the moisture is removed while maintaining the temperature of the low temperature location low. In order to remove the moisture while maintaining the temperature of the low temperature location low, the inert gas is locally supplied to the low temperature location. Specifically, the distributor **186** is used to locally supply the inert gas to the low temperature location.

(56) Subsequently, a detailed structure of the distributor **186** will be described with reference to FIGS. 3A and 3B. FIG. 3A schematically illustrates the distributor **186** when viewed from the robot **170** toward the wall **141b**, and FIG. 3B schematically illustrates a cross-section of the distributor

186 taken along the line A-A' in FIG. 3A.

(57) The distributor **186** is constituted mainly by a main body **186a** of a cylindrical shape. The inert gas supply pipe **181** is connected to the main body **186a**. A hole **186b** serving as an inert gas supply hole is provided on a side of the main body **186a** in the direction of the robot **170**. A hole **186c** serving as an inert gas supply hole may be further provided below the main body **186a**.

(58) A height of the hole **186b** in a height direction (vertical direction) is set such that the inert gas discharged (ejected) through the hole **186b** may collide with the ceiling **147**. For example, the hole **186b** is provided at a height position between the ceiling **147** and the arm **172** of the robot **170**. The arm **172** of the robot **170** is provided higher than the other arms of the robot **170**.

(59) The hole **186b** may be open in a direction in which the inert gas collides with an inner wall (that is, the wall **147a**) of the ceiling **147**. By supplying the inert gas toward the ceiling **147**, the moisture adhering to the wall **147a** of the ceiling **147** collides with the inert gas, so that the moisture can be physically peeled off. Therefore, it is possible to remove the moisture adhering to the wall **147a** of the ceiling **147** while maintaining the temperature of the wall **147a** or the robot **170** low.

(60) A width of the hole **186b** is set such that, for example, the inert gas can be supplied to the O-ring **152** in the horizontal direction. Specifically, the width of the hole **186b** is set to be equal to or greater than a diameter of the O-ring **152**. Thereby, the moisture adhering to the periphery of the O-ring **152** can collide with the inert gas, and the moisture can be physically peeled off. Therefore, it is possible to remove the moisture without thermally deforming the O-ring **152**. The present embodiment is described by way of an example in which the hole **186b** is configured as a single slit shape. However, the present embodiment is not limited thereto. For example, the hole **186b** may be configured as a plurality of holes. When the hole **186b** is configured as the plurality of the holes, a distance between outermost holes among the plurality of the hole is set to be equal to or greater than the diameter of the O-ring **152**.

(61) The hole **186c** is provided below the main body **186a**. The inert gas supplied through the hole **186c** is supplied toward the wall **141a** adjacent to the load lock chamber **130**. The reason for supplying the inert gas to the wall **141a** will be described below. The unprocessed substrates **100** stored in the pod **111** may be transferred to various places in a factory where the substrate processing apparatus **200** is installed. Therefore, the moisture may adhere to the unprocessed substrates **100** before the unprocessed substrates **100** are transferred to the substrate processing apparatus **200**. The moisture adhering to the unprocessed substrates **100** may be diffused into the vacuum transfer chamber **140** when the unprocessed substrates **100** are transferred to the vacuum transfer chamber **140** from the load lock chamber **130**. In particular, the moisture may tend to adhere to the wall **141a** arranged in the vicinity of the communication hole **142**.

(62) On the other hand, by supplying the inert gas supplied through the hole **186c** toward the wall **141a**, the moisture adhering to the wall **141a** may collide with the inert gas, and the moisture can be physically peeled off. Therefore, it is possible to efficiently remove a large amount of the moisture adhering to the wall **147a** while maintaining the temperature of the wall **147a** or the robot **170** low.

(63) Since the moisture adhering to the unprocessed substrates **100** passing through the load lock chamber **130** is diffused around the communication hole **142**, it is preferable that a width of the hole **186c** parallel to the wall **141a** is set to be equal to or greater than a width of the communication hole **142**. Therefore, by setting the width of the hole **186c** equal to the width of the communication hole **142**, it is possible to reliably supply the inert gas to the moisture adhering to a portion of the wall **141a** around the communication hole **142**. Further, by setting the width of the hole **186c** greater than the width of the communication hole **142**, it is possible to reliably supply the inert gas to the moisture adhering to a side portion of the wall **141a** around a side portion of the communication hole **142**.

(64) A maximum width of the hole **186c** in the direction parallel to the wall **141a** is set to be a

distance between the facing walls of the transfer chamber **140** adjacent to the wall **141a**. It is more preferable that the maximum width of the hole **186c** is set to be equal to a width of the wall **141a**.
(65) It is more preferable that the inert gas supplied through the distributor **186** is heated by the heater **185**. By heating the inert gas by the heater **185**, it is possible to increase an efficiency of removing the moisture. When the inert gas is supplied, the supply of the inert gas and the stop of the supply of the inert gas may be alternately and repeatedly performed. By repeatedly colliding the moisture with the inert gas, the moisture can be physically removed more efficiently.

(66) <Process Module>

(67) Subsequently, the process modules PM will be described mainly on the reactors RCs. Since the configurations of the process modules PM1 through PM4 are the same, the process module PM serving as one of the process modules PM1 through PM4 will be described below. Similarly, since the configurations of the reactors RC1 through RC8 are the same, the reactor RC serving as one of the reactors RC1 through RC8 will be described below.

(68) A partition wall is provided between the two reactors RCs provided in the process module PM. The partition wall is configured to prevent mixing of inner atmospheres of process spaces **205** of the two reactors RCs. That is, the inner atmospheres of the process spaces **205** are maintained independently.

(69) The reactor RC will be described in detail with reference to FIGS. **4** and **5**. Since the configurations of the two reactors RCs provided in the process module are the same, the reactor RC serving as one of the two reactors RCs will be described below. As shown in FIG. **4**, the reactor RC includes a vessel **202**. For example, the vessel **202** includes a flat and sealed vessel whose horizontal cross-section is circular. The vessel **202** is made of a metal material such as aluminum (Al) and stainless steel (SUS) or quartz. A process chamber **201** defining the process space **205** where the substrate **100** such as a silicon wafer is processed and a transfer chamber **206** defining a transfer space through which the substrate **100** is transferred into or out of the process space **205** are provided in the vessel **202**. The vessel **202** is constituted by an upper vessel **202a** and a lower vessel **202b**. A partition plate **208** is provided between the upper vessel **202a** and the lower vessel **202b**.

(70) The communication hole **148** is provided adjacent to the gate valve **149** at a side surface of the lower vessel **202b**. The substrate **100** is transferred between the transfer chamber **206** and the vacuum transfer chamber **140** through the communication hole **148**. Lift pins **207** are provided at a bottom of the lower vessel **202b**.

(71) A substrate support **210** configured to support the substrate **100** is provided in the process space **205**. The substrate support **210** mainly includes: the substrate mounting table **212** provided with a substrate placing surface **211** on which the substrate **100** is placed; and the heater **213** which is a heating structure embedded in the substrate mounting table **212**. Through-holes **214** through which the lift pins **207** penetrate are provided at positions of the substrate mounting table **212** corresponding to the lift pins **207**.

(72) Wiring **222** configured to supply the electric power to the heater **213** is connected to the heater **213**. The wiring **222** is connected to a heater controller **223**. The heater controller **223** is electrically connected to the controller **400**. The controller **400** is configured to control the heater controller **223** to operate the heater **213**.

(73) The substrate mounting table **212** is supported by a shaft **217**. The shaft **217** penetrates the bottom of the vessel **202**, and is connected to an elevator **218** at the outside of the vessel **202**.

(74) The substrate **100** placed on the substrate placing surface **211** of the substrate mounting table **212** may be elevated or lowered by operating the elevator **218** by elevating and lowering the shaft **217** and the substrate mounting table **212**.

(75) For example, the process chamber **201** is constituted by a buffer structure **230** described later and the substrate mounting table **212**. The process chamber **201** may be configured by another structure as long as the process space **205** in which the substrate **100** is processed can be secured.

(76) When the substrate **100** is transferred, the substrate mounting table **212** is lowered until the substrate placing surface **211** faces the communication hole **148**, that is, until a transfer position **P0** is reached. When the substrate **100** is processed, the substrate mounting table **212** is elevated until the substrate **100** reaches a processing position in the process space **205** as shown in FIG. **4**.

(77) The buffer structure **230** configured to diffuse a gas is provided in an upper portion (upstream side) of the process space **205**. The buffer structure **230** is constituted mainly by a lid **231**. A first gas supplier **240** and a second gas supplier **250**, which will be described later, are connected to the lid **231** so as to communicate with a gas introduction hole **231a** provided in the lid **231**. Although the gas introduction hole **231a** configured as a single hole is illustrated in FIG. **4**, holes serving as the gas introduction hole **231a** may be provided with respect to gas suppliers such as the first gas supplier **240** and the second gas supplier **250**, respectively.

(78) <Exhauster>

(79) Subsequently, an exhauster (which is an exhaust system) **271** will be described. An exhaust pipe **272** configured to communicate with the process space **205** is provided. The exhaust pipe **272** is connected to the upper vessel **202a** so as to communicate with the process space **205**. An APC **273** serving as a pressure controller (pressure regulator) is provided at the exhaust pipe **272**. The APC **273** is capable of controlling an inner pressure of the process space **205** to a predetermined pressure.

(80) The APC **273** includes a valve body (not shown) whose opening degree can be adjusted. The APC **273** can adjust the conductance of the exhaust pipe **272** in accordance with an instruction from the controller **400**. Further, a valve **274** is provided at the exhaust pipe **272** at an upstream side of the APC **273**. The exhaust pipe **272**, the APC **273** and the valve **274** may be collectively referred to as the exhauster **271**.

(81) Further, a dry pump DP **275** is provided at a downstream side of the exhaust pipe **272**. The dry pump **275** is capable of exhausting the inner atmosphere of the process space **205** through the exhaust pipe **272**.

(82) <Gas Supplier>

(83) Subsequently, a gas supplier (which is a gas supply system) capable of supplying a gas to the process chamber **201** will be described with reference to FIG. **5**. The gas supplier described with reference to FIG. **5** may also be referred to as a “process chamber gas supplier” in order to distinguish it from the transfer chamber inert gas supplier described above.

(84) The first gas supplier (which is a first gas supply system) **240** will be described. A first gas supply source **242**, a mass flow controller (MFC) **243** serving as a flow rate controller (flow rate control structure) and a valve **244** serving as an opening/closing valve are sequentially provided in order at a first gas supply pipe **241** from an upstream side toward a downstream side of the first gas supply pipe **241**.

(85) The first gas supply source **242** is a source of a first gas (also referred to as a “first element-containing gas”) containing the first element. The first element-containing gas serves as a source gas, which is one of process gases. In the present embodiment, for example, the first element includes silicon (Si). That is, for example, the first element-containing gas includes a silicon-containing gas. Specifically, a monosilane (SiH₄) gas may be used as the silicon-containing gas.

(86) The first gas supplier **240** is constituted mainly by the first gas supply pipe **241**, the MFC **243** and the valve **244**.

(87) Subsequently, the second gas supplier (which is a second gas supply system) **250** will be described. A second gas supply source **252**, a mass flow controller (MFC) **253** serving as a flow rate controller (flow rate control structure) and a valve **254** serving as an opening/closing valve are sequentially provided in order at a second gas supply pipe **251** from an upstream side toward a downstream side of the second gas supply pipe **251**.

(88) The second gas supply source **252** is a source of a second gas (also referred to as a “second

element-containing gas”) containing the second element. The second element-containing gas is one of the process gases. The second element-containing gas may serve as a reactive gas or a modifying gas.

(89) In the present embodiment, the second element-containing gas contains the second element different from the first element. For example, the second element includes an element selected from the group of oxygen (O), nitrogen (N) and carbon (C). The present embodiment will be described by way of an example in which an oxygen-containing gas is used as the second element-containing gas. Specifically, oxygen gas (O₂ gas) may be used as the oxygen-containing gas.

(90) The second gas supplier **250** is constituted mainly by the second gas supply pipe **251**, the MFC **253** and the valve **254**.

(91) When a film is formed on the substrate **100** by using the first gas alone, the second gas supplier **250** may not be provided in the substrate processing apparatus **200**.

(92) <Controller>

(93) Subsequently, the controller **400** will be described with reference to FIG. **6**. The substrate processing apparatus **200** includes the controller **400** configured to control operations of components constituting the substrate processing apparatus **200**.

(94) The controller **400**, which is a control apparatus (control structure) may be embodied by a computer including a CPU (Central Processing Unit) **401**, a RAM (Random Access Memory) **402**, a memory **403** serving as a storage and an I/O port (input/output port) **404**. The RAM **402**, the memory **403** and the I/O port **404** may exchange data with the CPU **401** via an internal bus **405**. The transmission/reception of the data in the substrate processing apparatus **200** may be performed by an instruction from a transmission/reception instruction controller **406**, which is also one of functions of the CPU **401**.

(95) The CPU **401** may further include a determination controller **407**. The determination controller **407** is capable of analyzing a relationship between a table stored in the memory **403** and the amount of the moisture measured by the moisture detector **146**.

(96) A network transmitter/receiver **283** connected to a host apparatus **270** via a network is provided. For example, the network transmitter/receiver **283** is capable of receiving information regarding a processing history and a processing schedule of the substrate **100** in the lot.

(97) The memory **403** may be embodied by a component such as a flash memory and a HDD (Hard Disk Drive). For example, a recipe **409** such as a process recipe in which information such as the sequences and the conditions of the substrate processing described later is stored are readably stored in the memory **403** and a control program **410** for controlling the operation of the substrate processing apparatus **200** is stored may be readably stored in the memory **403**. The memory **403** may further include a moisture information memory **411** capable of recording the data detected by the moisture detector **146** and reading temperature data thereof.

(98) The process recipe is a program that is executed by the controller **400** to obtain a predetermined result by performing the sequences of the substrate processing described later. Hereinafter, the process recipe and the control program may be collectively or individually referred to simply as a “program.” In the present specification, the term “program” may refer to the process recipe alone, may refer to the control program alone, or may refer to both of the process recipe and the control program. The RAM **402** serves as a memory area (work area) in which the program or the data read by the CPU **401** are temporarily stored.

(99) The I/O port **404** is electrically connected to the components of the process modules PMs described above such as the gate valve **149**, the elevator **218**, the pressure regulators, the pumps and the heater controller **223**.

(100) The CPU **401** is configured to read and execute the control program from the memory **403** and read the process recipe in accordance with an instruction such as an operation command inputted from an input/output device **281**. The CPU **401** is configured to control various operations in accordance with the process recipe such as an opening and closing operation of the gate valve

149, an elevating and lowering operation of the elevator **218**, an operation of the moisture detector **146**, an operation of the heater controller **223**, an ON/OFF control operation of each pump, a flow rate adjusting operation of each MFC described above and an operation of each valve described above.

(101) For example, the controller **400** according to the present embodiment may be embodied by preparing an external memory **282** (for example, a magnetic disk such as a hard disk, an optical disk such as a DVD, a magneto-optical disk such as an MO, a semiconductor memory such as a USB memory) storing the program described above and installing the program onto a computer using the external memory **282**. The method of providing the program to the computer is not limited to the external memory **282**. For example, the program may be directly provided to the computer by a communication means such as the Internet and a dedicated line instead of the external memory **282**. The memory **403** and the external memory **282** may be embodied by a non-transitory computer-readable recording medium. Hereinafter, the memory **403** and the external memory **282** are collectively or individually referred to as a recording medium. In the present specification, the term "recording medium" may refer to the memory **403** alone, may refer to the external memory **282** alone, or may refer to both of the memory **403** and the external memory **282**.

(102) (2) Substrate Processing

(103) Hereinafter, as an example of the substrate processing performed by the substrate processing apparatus **200**, a film processing step (also referred to as a film-forming process) of forming a film on the substrate **100** by using the substrate processing apparatus **200** described above and a maintenance step will be described. In the following description, the controller **400** controls the operations of the components constituting the substrate processing apparatus **200**.

(104) The substrate processing will be described by way of an example in which the vacuum transfer chamber **140** and the reactor RC among the reactors RCs are used.

(105) <Film Processing Step>

(106) A substrate transfer step, which is a part of one of the film processing step, will be described. The atmospheric transfer robot **122** transfers the substrate **100** out of the pod **111**. Then, the atmospheric transfer robot **122** transfers the substrate **100** to the load lock chamber **130**. When transferring the substrate **100** to the load lock chamber **130**, if there is the processed substrate **100** in the load lock chamber **130**, the atmospheric transfer robot **122** transfers the processed substrate **100** to the pod **111**.

(107) An inner atmosphere (inner pressure) of the load lock chamber **130** is adjusted to a negative pressure. When the inner pressure of the load lock chamber **130** is adjusted to the same pressure level as the inner pressure of the vacuum transfer chamber **140**, the gate valve **134** is opened. The robot **170** picks up the unprocessed substrates **100** in the load lock chamber **130** and transfers it to each of the reactors RCs. When transferring the unprocessed substrates **100** to each of the reactors RCs, the moisture adhering to the unprocessed substrates **100** may be diffused into the vacuum transfer chamber **140**.

(108) After processing the substrates **100** for a predetermined time in each of the reactors RCs, the gate valve **149** is opened. Then, the robot **170** replaces the processed substrate **100** in the reactor RC with the unprocessed substrate **100** supported by the robot **170**, and transfers (loads) the unprocessed substrate **100** into the reactor RC.

(109) The robot **170** transfers (loads) the processed substrate **100** into the load lock chamber **130**.

(110) While performing the operations described above, the moisture detector **146** detects the amount of the moisture in the vacuum transfer chamber **140**. When the amount of the moisture is equal to or greater a predetermined value, the maintenance step is performed before the next substrate are processed or before the substrates of the next lot are processed. When the amount of the moisture is less than the predetermined value, the substrates **100** are continuously processed.

(111) Subsequently, the operation in the reactor RC when processing the substrate **100** will be described.

(112) The substrate mounting table **212** is lowered to a position of transferring the substrate **100** (that is, the transfer position **P0**) such that the lift pins **207** penetrate through the through-holes **214** of the substrate mounting table **212**. As a result, the lift pins **207** protrude from the surface of the substrate mounting table **212** by a predetermined height. In parallel with the operation of lowering the substrate mounting table **212**, an inner atmosphere of the transfer chamber **206** is exhausted such that an inner pressure of the transfer chamber **206** is the same as that of the vacuum transfer chamber **140** provided adjacently or lower than that of the vacuum transfer chamber **140** provided adjacently.

(113) Subsequently, the gate valve **149** is opened to communicate the transfer chamber **206** with the vacuum transfer chamber **140** provided adjacently. Then, the robot **170** loads the substrate **100** from the vacuum transfer chamber **140** into the transfer chamber **206** and places the substrate **100** on the lift pins **207**.

(114) After the substrate **100** is placed on the lift pins **207**, the substrate mounting table **212** is elevated until the substrate **100** is placed on the substrate placing surface **211**. Then, the substrate mounting table **212** is further elevated until the substrate **100** reaches the processing position as shown in FIG. 4.

(115) When the substrate **100** is being placed on the substrate placing surface **211**, the electric power is supplied to the heater **213** such that a temperature of the surface of the substrate **100** is adjusted to a predetermined temperature. For example, the temperature of the surface of the substrate **100** is adjusted to the predetermined temperature equal to or greater than the room temperature and equal to or less than 800° C., preferably, equal to or greater than the room temperature and equal to or less than 500° C. When heating the substrate **100** to the predetermined temperature, the wall **141b** is also heated.

(116) Subsequently, a process gas supply step, which is a part of one of the film processing step, will be described. When the substrate **100** is heated and reaches a desired temperature, the first gas and the second gas are supplied to the process chamber **201**. As a method of supplying the first gas and the second gas, for example, the first gas and the second gas are supplied simultaneously or supplied alternately to form a desired film. In the present embodiment, for example, the desired film may refer to a silicon oxide film.

(117) When the desired film is formed on the substrate **100**, the substrate **100** is transferred (unloaded) out of the process chamber **201** in the order reverse to that of the loading of the substrate **100** described above. Since the amount of the moisture in the transfer chamber **140** is less than predetermined value described above, it is possible to suppress the reduction of the yield.

(118) <Maintenance Step>

(119) Subsequently, the maintenance step will be described. When the determination controller **407** determines that the amount of the moisture detected by the moisture detector **146** is equal to or greater than the predetermined value described above, the maintenance step is performed. The maintenance step is performed while the substrate **100** is not present in the transfer chamber **140** and the operation related to the processing of the substrate **100** is stopped. For example, operations such as an operation of supplying the gas to the process chamber **201** and an operation of transferring the substrate **100** are stopped when the maintenance step is performed.

(120) In the maintenance step, the inert gas supplier **180** and the exhaustor **160** are operated. By supplying the inert gas into the housing **141**, the moisture adhering to the low temperature location of the housing **141** is removed. According to the present embodiment, the moisture adhering to the wall **147a** is removed by supplying the inert gas through the hole **186b** of the distributor **186** toward the wall **147a** which serves as the low temperature location. Specifically, the inert gas is supplied to the wall **147a** of the ceiling **147**, which constitutes the low temperature region. After a predetermined time has elapsed, the supply of the inert gas is stopped.

(121) When the distributor **186** is provided with the hole **186c**, the inert gas may be supplied in the direction of the wall **141a** to remove the moisture adhering to the wall **141a**.

(122) In the maintenance step, a supply amount of the inert gas may be controlled based on the information on the amount of the moisture detected by the moisture detector **146**. For example, when the determination controller **407** determines that the amount of the moisture detected by the moisture detector **146** is greater than the predetermined value, it is determined that the amount of the moisture adhering to the low temperature location is large, and the supply amount of the inert gas may be increased. Thereby, it is possible to reliably remove the moisture.

(123) For example, when the determination controller **407** determines that the amount of the moisture detected by the moisture detector **146** is lower than the predetermined value, it is determined that the amount of the moisture adhering to the low temperature location is small, and the supply amount of the inert gas may be decreased. Thereby, it is possible to suppress the supply amount of the inert gas, and it is also possible to reduce the cost related to the amount of the inert gas used in the maintenance step.

(124) When controlling the supply amount of the inert gas based on the information of the amount of the moisture detected by the moisture detector **146**, for example, the moisture information memory **411** may store in advance a table in which the amount of the moisture and the supply amount of the inert gas are related. In such a case, the determination controller **407** may determine the supply amount of the inert gas by comparing the amount of the moisture detected by the moisture detector **146** and the table stored in advance in the moisture information memory **411**.

(125) According to the present embodiment, the supply of the inert gas is stopped after a predetermined time has elapsed. However, the technique of the present disclosure is not limited thereto. For example, the supply of the inert gas may be stopped when it is determined that the amount of the moisture detected by the moisture detector **146** is equal to or less than the predetermined value.

Second Embodiment

(126) Subsequently, a second embodiment according to the technique of the present disclosure will be described with reference to FIGS. **7** and **8**. A structure of the distributor **186** of the second embodiment is different from that of the first embodiment. In the present embodiment, the distributor **186** is further provided with a nozzle **187** which is an elongated structure. Hereinafter, the distributor **186** and the nozzle **187** of the second embodiment will be mainly described. Since other configurations of the second embodiment are the same as those of the first embodiment, the description thereof will be omitted. Further, the low temperature region (second region) in the present embodiment may refer to a central region of the wall **147a**, which will be described later.

(127) The distributor **186** in the present embodiment is provided with the nozzle **187** instead of providing the hole **186b** shown in FIG. **3**. The nozzle **187** communicates with the inert gas supply pipe **181** via the distributor **186**. The nozzle **187** is provided with a hole **187a** through which the inert gas is ejected (discharged). The hole **187a** is open toward the wall **147a**. The nozzle **187** extends along the ceiling **147**.

(128) The nozzle **187** is configured such that the inert gas can be supplied toward at least a central portion (that is, the central region) of the wall **147a**. As described above, the central portion of the wall **147a** is separated from each of the reactors RCs. Therefore, a temperature of the central portion of the wall **147a** tends to be lowered and the moisture may easily adhere to the central portion of the wall **147a**. However, with the configuration such as the nozzle **187** described above, it is possible to reliably supply the inert gas to the center of the wall **147a**. Thereby, it is possible to remove the moisture adhering to the central portion of the wall **147a**.

Third Embodiment

(129) Subsequently, a third embodiment according to the technique of the present disclosure will be described with reference to FIG. **9**. A structure of the distributor **186** of the third embodiment is different from that of the first embodiment. Opening directions of holes in the distributor **186** of the third embodiment are different from that in the distributor **186** of the first embodiment. Hereinafter, the distributor **186** of the third embodiment will be mainly described. Since other configurations of

the third embodiment are the same as those of the first embodiment, the description thereof will be omitted. Further, the low temperature region (second region) in the present embodiment may refer to a region constituted by walls interposed between a plurality of process chambers including the process chamber **201** or walls interposed between the load lock chamber **130** and the plurality of the process chambers including the process chamber **201** as described later.

(130) As shown in FIG. **9**, the distributor **186** of the present embodiment is provided with holes **186d**, **186e**, **186f**, **186g** and **186h**. Subsequently, opening direction of the holes **186d** through **186h** will be described. The holes **186d** through **186h** are opened so that the inert gas can be supplied to the low temperature location such as a wall **191**, a wall **192**, a wall **193**, a wall **194** and a wall **195**, as described by arrows shown in FIG. **9**.

(131) Subsequently, the wall **191**, the wall **192**, the wall **193**, the wall **194** and the wall **195** will be described. As described above, the temperature of the housing **141** in the vicinity of the reactor RC is high due to the influence of the heater **213**. In particular, the temperature in the vicinity of the communication hole **148** configured to communicate with the reactor RC and the housing **141** or the temperature of the wall **141b** at which the communication hole **148** is provided may be high. However, the wall **191**, the wall **192**, the wall **193**, the wall **194** and the wall **195**, which are provided between the adjacent reactors RCs or between the reactor RC and the load lock chamber **130** are less affected by the heater **213**. As a result, the temperatures of the walls **191** through **195** are lower than the temperature in the vicinity of the communication hole **148**. In particular, when the chiller or the cooling water is flowing through the housing **141**, the temperatures of the walls **191** through **195** are further lowered. Then, the moisture may easily adhere to the walls **191** through **195**.

(132) Therefore, according to the present embodiment, a structure is provided such that the inert gas can be supplied to the walls interposed between the adjacent reactors RCs or between the reactor RC and the load lock chamber **130** (that is, the walls **191** through **195**) so as to remove the moisture adhering to the walls **191** through **195**.

(133) Specifically, for the wall **191** adjacent to the load lock chamber **130** and the reactor RC1, the hole **186d** is configured to face the wall **191** so that the inert gas can be supplied to the wall **191**. For the wall **192** adjacent to the reactor RC2 and the reactor RC3, the hole **186e** is configured to face the wall **192** so that the inert gas can be supplied to the wall **192**. For the wall **193** adjacent to the reactor RC4 and the reactor RC5, the hole **186f** is configured to face the wall **193** so that the inert gas can be supplied to the wall **193**. For the wall **194** adjacent to the reactor RC6 and the reactor RC7, the hole **186g** is configured to face the wall **194** so that the inert gas can be supplied to the wall **194**. For the wall **195** adjacent to the load lock chamber **130** and the reactor RC8, the hole **186h** is configured to face the wall **195** so that the inert gas can be supplied to the wall **195**.

(134) In the distributor **186** according to the present embodiment, surfaces facing the wall **191**, the wall **192**, the wall **193**, the wall **194** and the wall **195** are provided so that the holes **186d** through **186h** can be configured, and each of the holes **186d** through **186h** is provided on the surfaces. With such a configuration, it is possible to reliably remove the moisture adhering to the walls interposed between the adjacent reactors RCs or between the reactor RC and the load lock chamber **130** (that is, the walls **191** through **195**). Since the wall **191**, the wall **192**, the wall **193**, the wall **194** and wall **195** are arranged between the communication holes, the walls **191** through **195** may also be referred to as “walls between the communication holes”.

Fourth Embodiment

(135) Subsequently, a fourth embodiment according to the technique of the present disclosure will be described with reference to FIGS. **10** and **11**. A structure of the distributor **186** of the fourth embodiment is different from that of the third embodiment. In the present embodiment, the distributor **186** is further provided with a plurality of nozzles (for example, a nozzle **188-1**, a nozzle **188-2** and a nozzle **188-3**) which is an elongated structure. The nozzles **188-1** through **188-3** may be collectively referred to as nozzles **188**. Hereinafter, the distributor **186** and the nozzles **188** of

the fourth embodiment will be mainly described. Since other configurations of the fourth embodiment are the same as those of the third embodiment, the description thereof will be omitted. Further, the low temperature region (second region) in the present embodiment may refer to the region constituted by the walls interposed between the plurality of the process chambers including the process chamber **201** or the walls interposed between the load lock chamber **130** and the plurality of the process chambers including the process chamber **201** as described later.

(136) Similar to the third embodiment, the distributor **186** of the present embodiment is provided with the hole **186d** and the hole **186h**. Similar to the third embodiment, the hole **186d** is configured to supply the inert gas toward the wall **191**, and the hole **186h** is configured to supply the inert gas toward the wall **195**.

(137) Instead of the hole **186e** of the third embodiment, the nozzle **188-1** is provided. Instead of the hole **186f** of the third embodiment, the nozzle **188-2** is provided. Instead of the hole **186g** of the third embodiment, the nozzle **188-3** is provided. As shown in FIG. **11**, a plurality of holes **188b** are provided at tips (front ends) of the nozzles **188**, respectively. The plurality of the holes **188b** are arranged in the vicinity of the wall **192**, the wall **193** and the wall **194**. The plurality of the holes **188b** are configured to supply the inert gas toward the wall **192**, the wall **193** and the wall **194**.

(138) With such a configuration, it is possible to reliably transfer the inert gas to the wall **192**, the wall **193** and the wall **194**. Therefore, it is possible to more reliably remove the moisture adhering to the low temperature location between the adjacent reactors RCs.

(139) Further, as shown in FIG. **11**, in each nozzle **188**, a plurality of holes **188a** may be provided between a hole among the plurality of the holes **188b** and the distributor **186**. Similar to the hole **187a** shown in FIG. **8**, the plurality of the holes **188a** are configured to supply the inert gas toward the wall **147a**. With such a configuration, it is possible to remove the moisture adhering to the wall **147a**.

(140) According to the present embodiment, the inert gas is transferred (supplied) from the wall **192** to the wall **194** using the nozzles **188**. However, the technique of the present disclosure is not limited thereto. For example, the inert gas may be directly supplied to each space from the wall **147a**. For example, the ceiling **147** may be provided with an inert gas supply hole capable of supplying the inert gas in an upper portion of each space, and the inert gas may be supplied to each space from the inert gas supply hole.

Other Embodiments

(141) While the technique is described in detail by way of the embodiments, the above-described technique is not limited thereto. The above-described technique may be modified in various ways without departing from the gist thereof.

(142) For example, the above-described embodiments are described by way of an example in which the film is formed by using the monosilane gas as the first element-containing gas (first gas) and using the O.sub.2 gas as the second element-containing gas (second gas) in the film-forming process performed by the substrate processing apparatus **200**. However, the above-described technique is not limited thereto. For example, the above-described technique may also be applied to film-forming processes wherein other gases are used to form different films.

(143) While the embodiments are described by way of an example in which the film-forming process is performed using two types of gases, that is the first gas and the second gas, the above-described technique is not limited thereto. For example, the above-described technique may also be applied to film-forming processes using one type of gas alone or three or more types of gases.

(144) As described above, according to some embodiments in the present disclosure, it is possible to reduce the amount of the moisture in the low temperature region in the substrate processing apparatus provided with the transfer chamber.

Claims

1. A substrate processing apparatus comprising: a process chamber provided with a heater; a load lock chamber; a transfer chamber provided between the process chamber and the load lock chamber and comprising a first region provided adjacent to the process chamber and a second region provided more adjacent to the load lock chamber than the first region and whose temperature is lower than a temperature of the first region, wherein the second region is a ceiling or a part of a sidewall of the transfer chamber; and an inert gas supplier comprising a supply hole capable of supplying an inert gas toward the second region in the transfer chamber while a substrate is not present in the transfer chamber.
2. The substrate processing apparatus of claim 1, further comprising: a window provided at the ceiling of the transfer chamber; and a seal provided between the window and a wall constituting the ceiling, wherein the second region comprises the ceiling, and the inert gas is capable of being supplied through the supply hole toward the wall constituting the ceiling.
3. The substrate processing apparatus of claim 1, further comprising: a window provided at the ceiling of the transfer chamber; and a seal provided between the window and a wall constituting the ceiling, wherein the seal is provided in the second region, and the inert gas is capable of being supplied through the supply hole toward the seal.
4. The substrate processing apparatus of claim 1, wherein the inert gas supplier further comprises an elongated structure configured to communicate with the inert gas supplier and extending along the ceiling of the transfer chamber, wherein the elongated structure is provided with the supply hole through which the inert gas is capable of being supplied.
5. The substrate processing apparatus of claim 1, wherein the inert gas supplier further comprises an elongated structure configured to communicate with the inert gas supplier and extending along the ceiling of the transfer chamber, wherein the elongated structure is provided with the supply hole through which the inert gas is capable of being supplied toward the ceiling.
6. The substrate processing apparatus of claim 1, wherein the inert gas supplier further comprises an elongated structure configured to communicate with the inert gas supplier and extending along the ceiling of the transfer chamber, wherein the second region comprises a central region of the ceiling.
7. The substrate processing apparatus of claim 1, wherein the transfer chamber comprises sidewalls comprising the sidewall provided adjacent to the load lock chamber among the sidewalls, and the second region comprises the part of the sidewall.
8. The substrate processing apparatus of claim 1, wherein the second region comprises a communication hole installed on the sidewall and configured to communicate with the load lock chamber and the transfer chamber, and the inert gas supplier is provided with the supply hole through which the inert gas is capable of being supplied toward the communication hole.
9. The substrate processing apparatus of claim 8, wherein a width of the supply hole is greater than a width of the communication hole.
10. The substrate processing apparatus of claim 9, wherein the process chamber comprises a plurality of process chambers, and the second region comprises sidewalls interposed between the plurality of process chambers or sidewalls interposed between the load lock chamber and the process chamber.
11. The substrate processing apparatus of claim 1, wherein the second region comprises sidewalls between a plurality of communication holes configured to communicate with the transfer chamber and the process chamber.
12. The substrate processing apparatus of claim 1, wherein the second region comprises sidewalls interposed between a plurality of communication holes configured to communicate with the transfer chamber and the process chamber, the supply hole is provided at an upper portion of the sidewalls interposed between the plurality of communication holes, and the inert gas is capable of being supplied through the communication holes toward the sidewalls interposed between the

plurality of communication holes.

13. The substrate processing apparatus of claim 1, wherein the inert gas supplier further comprises an elongated structure configured to communicate with the inert gas supplier and extending toward the second region, wherein the process chamber comprises a plurality of process chambers, the second region comprises sidewalls interposed between the plurality of process chambers or sidewalls interposed between the load lock chamber and the process chamber, and the elongated structure is capable of supplying the inert gas to the second region.

14. The substrate processing apparatus of claim 1, wherein the inert gas supplier comprises a heater capable of heating the inert gas.

15. The substrate processing apparatus of claim 1, wherein the inert gas supplier is controlled such that a supply of the inert gas and a stop of the supply of the inert gas are alternately and repeatedly performed.

16. The substrate processing apparatus of claim 1, wherein the inert gas supplier is configured to supply the inert gas to the transfer chamber while the substrate is not present in the transfer chamber.

17. The substrate processing apparatus of claim 1, wherein the inert gas supplier is configured to start supplying the inert gas to the transfer chamber and to stop supplying the inert gas when an amount of moisture detected by a detector is equal to or less than a predetermined value after a supply of the inert gas is started.

18. The substrate processing apparatus of claim 1, wherein a medium capable of adjusting a temperature is supplied to the sidewall of the transfer chamber.

19. A method of manufacturing a semiconductor device, comprising: processing the substrate by using the substrate processing apparatus of claim 1.

20. A non-transitory computer-readable recording medium storing a program that when executed by a processor of the substrate processing apparatus causes the substrate processing apparatus to perform the method of claim 19.
